OE10 — Conceptual Model of Property-Grounded CUT Feasibility

1. Modeling Scope and Design Rationale

The conceptual model represents a property-grounded approach to evaluating the feasibility of the CUT action in controlled kitchen scenarios, with potential applications to embodied AI and robotic task planning. Rather than relying on object names, categories, or conventional functions, the ontology evaluates candidate–target configurations using explicitly represented physical features, qualitative feature levels, relevant part structures, and target-dependent action requirements. An Assessment connects a candidate, target, and action, while ActionRequirement and RequirementMapping specify the physical conditions relevant to CUT feasibility. The current model assumes a reference performer with human-scale manipulation capabilities, without explicitly modeling robot-specific control or physical execution.

Assignment 10 builds on the existing model without fundamentally changing its conceptual structure. Additional controlled test cases, including a spatula with high rigidity and explicitly unspecified edge thickness, demonstrate how the existing representation and reasoning framework can be reused to evaluate different physical configurations and information states. The qualitative FeatureLevels (Low, Medium, and High) are interpreted within the controlled evaluation context. Future extensions may introduce feature-specific numerical criteria and broader data integration to support scalability, while preserving the current property-grounded reasoning approach.

2. Diagram Conventions and Interpretation

The conceptual model is presented through complementary class-structure and object-property views. In the class hierarchy diagrams, an open triangular arrowhead points from a subclass toward its superclass. Figures 1–4 present explicitly asserted subclass relationships, whereas Figure 5 illustrates selected subclass relationships that follow logically from OWL equivalent-class definitions. These two forms of classification should be distinguished. An asserted subclass relationship explicitly establishes that one class is included in another. An OWL equivalent-class definition specifies necessary and sufficient conditions for class membership, allowing a reasoner to classify individuals when the required conditions are established.

In the object-property diagram, directed arrows connect domain and range classes, with labels indicating the corresponding properties. Cardinality annotations are included where they reflect implemented OWL restrictions. The diagrams represent the conceptual and logical organization of the ontology; SWRL-based feasibility judgments are explained separately from OWL class classification.

3. Class Structure and OWL-Defined Classification

3.1 Asserted Class Hierarchies

The ontology organizes its main concepts into several class hierarchies, separating physical entities, physical features, action requirements, and feasibility judgments. **PhysicalEntity** distinguishes complete objects and their constituent parts through **WholeObject** and **ObjectPart**. **PhysicalFeature** organizes the characteristics used to describe candidates and targets, including geometric, mechanical, material, surface, and structural features. The corresponding **ActionRequirement** hierarchy represents the physical characteristics relevant to determining whether CUT requirements can be met. Although **PhysicalFeature** and **ActionRequirement** contain related categories, they serve different purposes. **PhysicalFeature** describes what is represented about an entity, whereas **ActionRequirement** describes what is needed for an action. Keeping these concepts separate allows candidate and target characteristics to be evaluated against action-specific requirements. The **FeasibilityJudgment** hierarchy distinguishes **Feasible**, **Infeasible**, and **InsufficientInformation** outcomes. Additional asserted subclass relationships include **TwoCUTRequirementsSatisfiedAssessment**, a specialization of **Assessment**, and **HighHardnessRigidityMapping**, a specialization of **RequirementMapping**. These classes support the representation and evaluation of specific conditions within the CUT model.

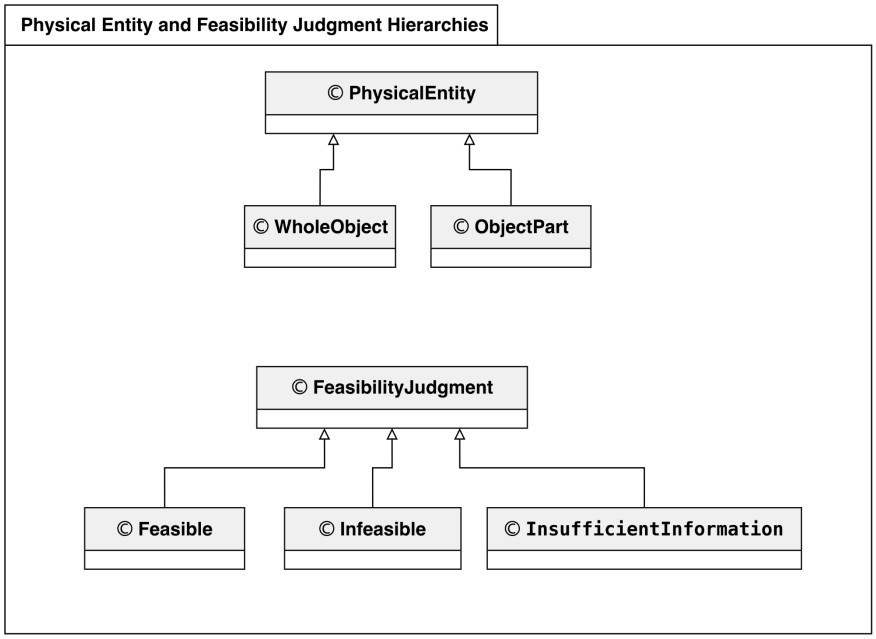


Figure 1. Physical Entity and Feasibility Judgment Hierarchies.

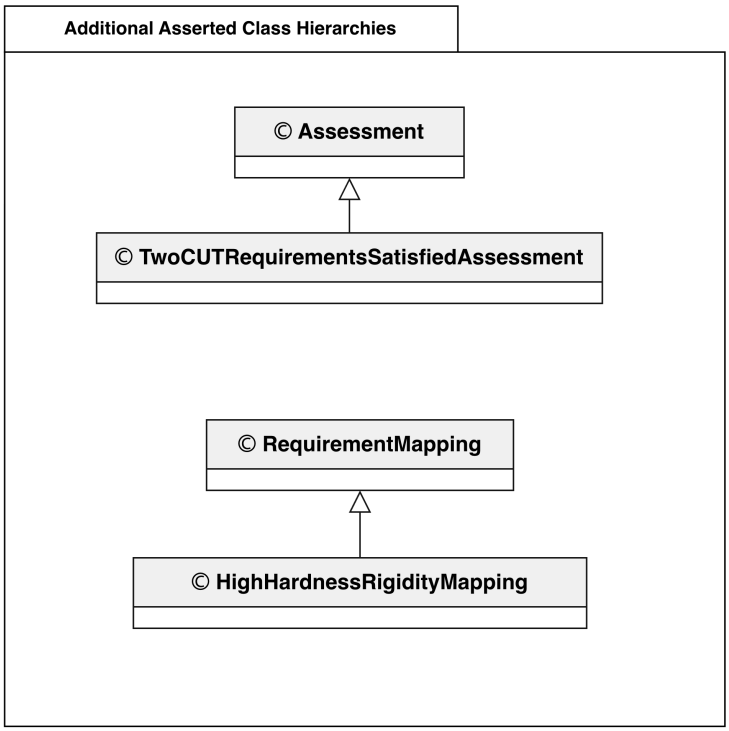


Figure 2. Additional Asserted Class Hierarchies.

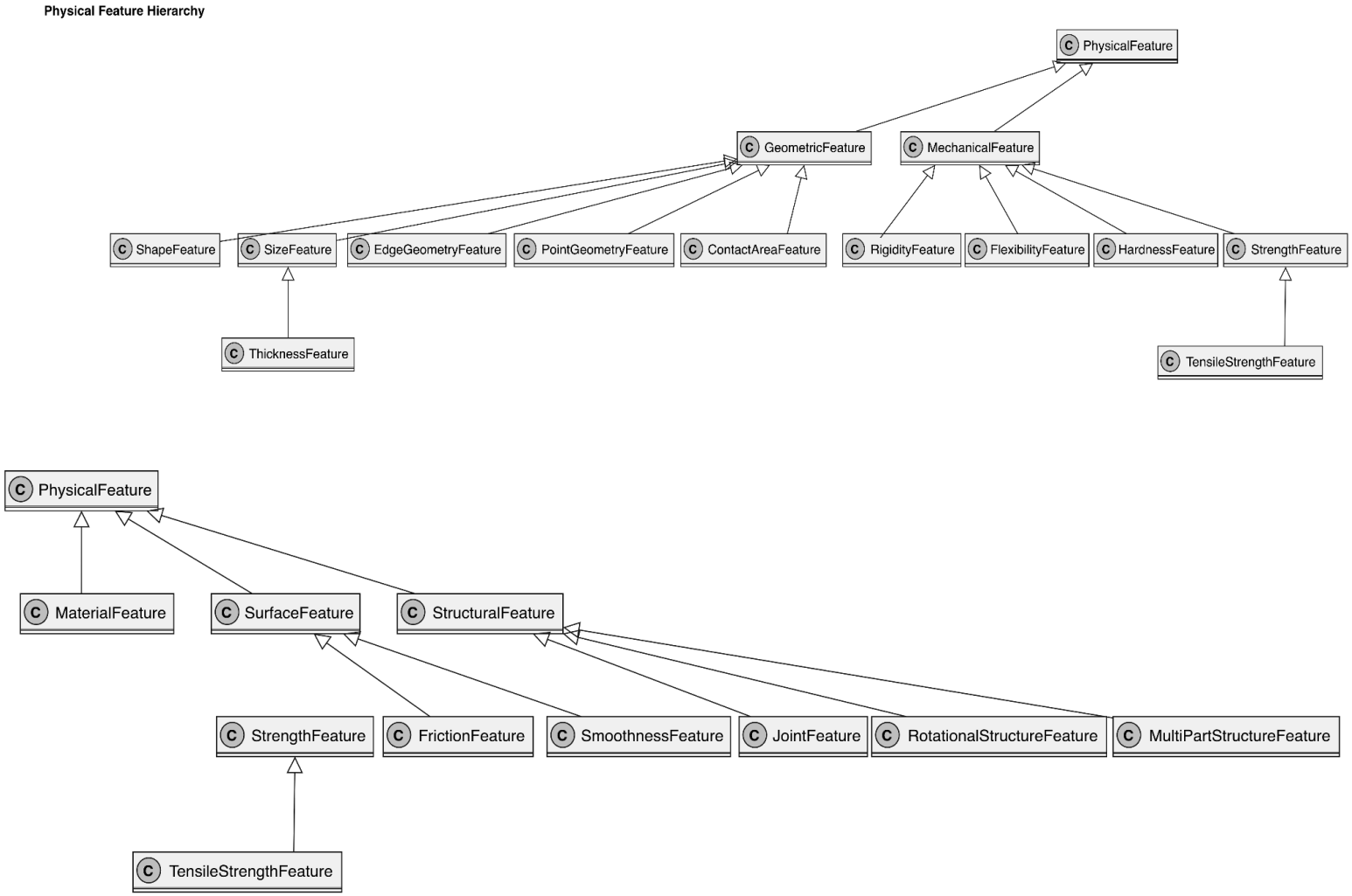


Figure 3. Physical Feature Hierarchy

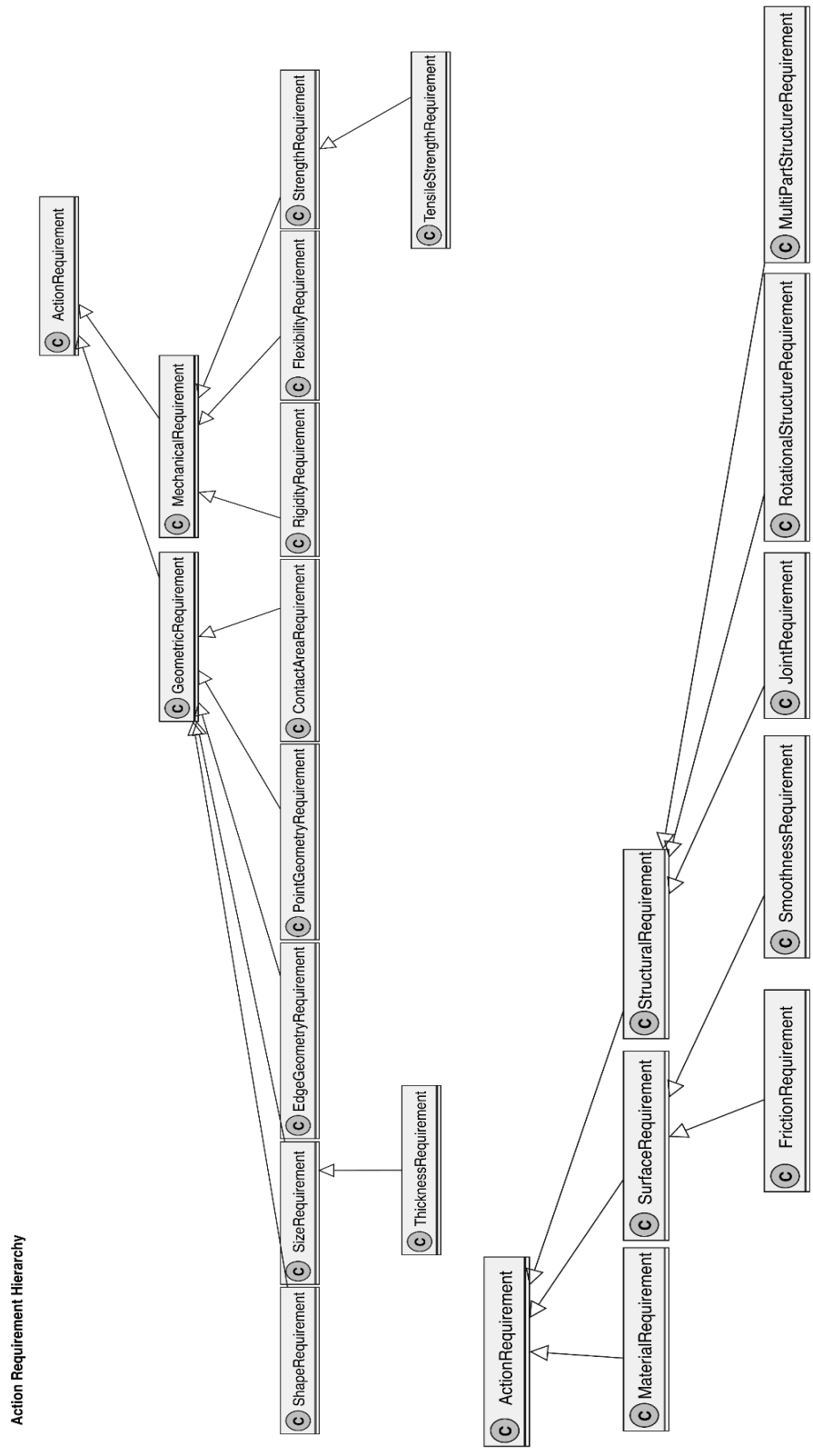


Figure 4. Action Requirement Hierarchy.

3.2 OWL-Defined Classification Patterns

In addition to explicitly asserted subclass relationships, the ontology includes classes defined through OWL equivalence axioms. These definitions allow class membership to be determined from represented relationships and restrictions rather than only from manually assigned class types. For example, **InteractingPart** is defined as an **ObjectPart** that interacts with at least one other **ObjectPart**. Consequently, an individual satisfying these conditions can be classified as **InteractingPart** through OWL reasoning. Similarly, **PartBearingObject**, **SinglePartObject**, **MultiPartObject**, **InteractingPartObject**, and **PairedInteractingObject** describe whole objects through restrictions on their part structures. **DescribedObject** captures whole objects with represented physical descriptions or parts. These patterns are particularly relevant to the ontology's treatment of part-based configurations. They provide a formal way to identify structural characteristics from explicit relationships, supporting the representation of configurations such as those used in the alternative-candidate CUT assessments.

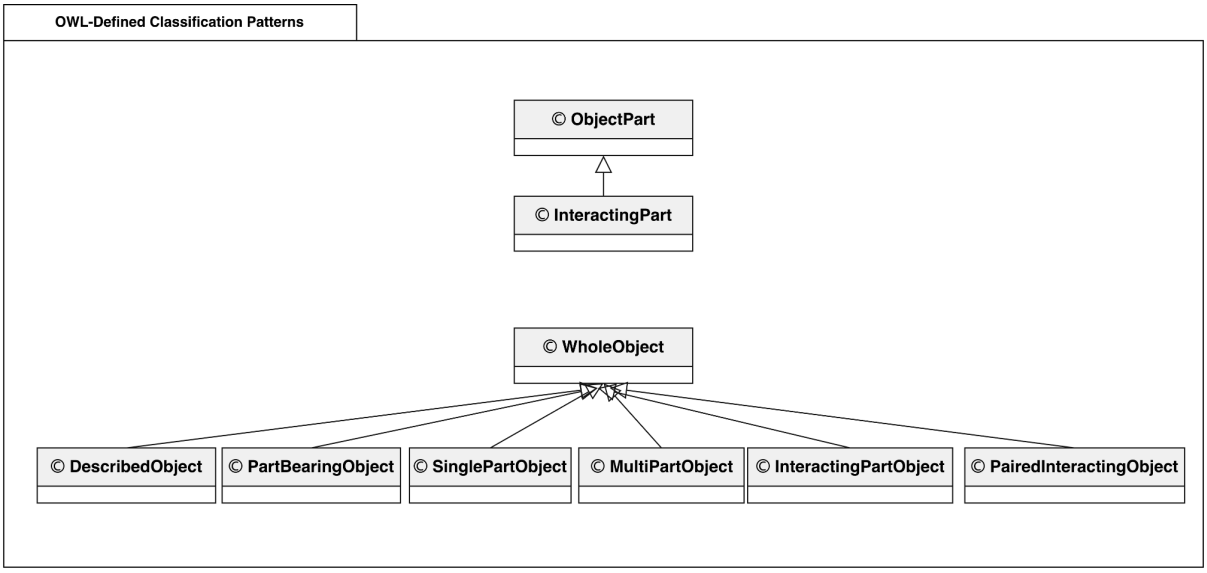


Figure 5. Selected OWL-Defined Classification Patterns.

Note: The subclass relationships shown in Figure 5 are entailed by OWL equivalent-class definitions rather than directly asserted through rdfs:subClassOf. These definitions specify necessary and sufficient conditions for class membership. The diagram presents selected classification relationships rather than the complete inferred hierarchy.

The distinction between Figures 1-4 and 5 highlights two complementary aspects of the ontology: explicit conceptual organization and logically defined classification. OWL reasoning supports structural classification, while the SWRL rules used for CUT assessment support the evaluation of action requirements and feasibility judgments.

4. Property-Grounded CUT Assessment Model

The central assessment model connects an **Assessment** to its candidate, target, action, and feasibility judgment through the properties *hasCandidate, hasTarget, hasAction*, and *hasJudgment*. Candidate and target are modeled as **PhysicalEntity** instances, with their roles determined by their relationships to the assessment rather than by separate object categories. Physical characteristics are represented using **FeatureValue**, which connects a **PhysicalFeature** to a categorical **FeatureLevel**. This structure allows an entity to carry explicit information about its physical characteristics without deriving those characteristics from its name or conventional function. An **Action** is linked to its **ActionRequirement** instances through *hasRequirement*, and each requirement identifies the relevant feature through *concernsFeature*.
RequirementMapping connects target feature conditions to the levels required for the corresponding action requirement. This allows CUT evaluation to account for both candidate characteristics and relevant target conditions. The model also uses *hasUnspecifiedFeatureType* to explicitly represent a feature type for which a value has not been supplied. This is important for distinguishing insufficient information from an explicitly conflicting feature value. Missing assertions alone are not treated as evidence of infeasibility.

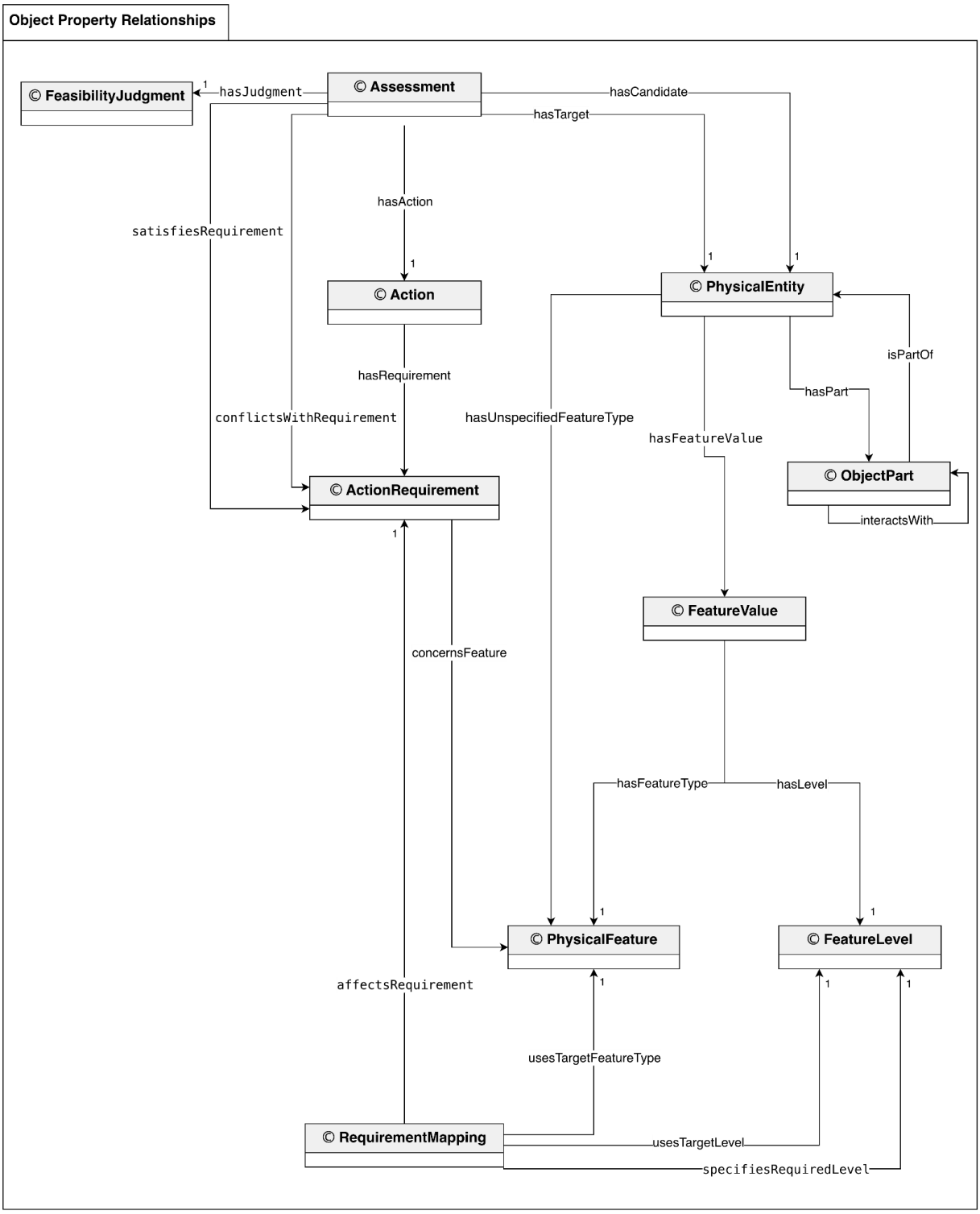


Figure 6. Object-Property Relationships Supporting Property-Grounded CUT Assessment.

Together, these relationships provide the structural basis for evaluating whether a configuration satisfies or conflicts with the modeled CUT requirements. SWRL rules can use these represented

conditions to infer requirement-evaluation relationships and feasibility judgments. The inferred results must be made available to the query environment when they are retrieved through SPARQL.

5. Part-Based Configuration Model

Some CUT configurations depend not only on whole-object characteristics but also on the physical features and relationships of their constituent parts. As shown in Figure 6, the ontology represents part–whole relationships through *hasPart* and *isPartOf*, and interactions between object parts through *interactsWith*. Because **ObjectPart** is a subclass of **PhysicalEntity**, individual parts can also have their own **FeatureValue** descriptions. These relationships support the representation and evaluation of alternative CUT configurations, including knife and scissors assessments, without using conventional object functions as reasoning evidence. The OWL-defined classes shown in Figure 5 further support structural classification based on explicitly represented part relationships.

6. Alignment with Competency Questions

The controlled assessment cases developed across Assignments 9 and 10 support the evaluation of different feasibility judgments. Test A, Test B, and Test C demonstrate satisfied requirements, conflicting requirements, and explicitly unspecified physical features. The additional spatula case further illustrates how the same ontology structure can represent incomplete information in a realistic kitchen scenario. Together, these cases demonstrate that feasibility reasoning is grounded in represented physical features, relevant part relationships, and requirement mappings rather than object names or conventional functions. The resulting judgments are interpreted within the modeled CUT requirements and evaluation conditions.

Competency Question	Relevant Model Components	Evaluation Focus
CQ1 — Single Configuration Feasibility	Assessment, FeatureValue, ActionRequirement, RequirementMapping	Evaluate one candidate–target configuration and explain the judgment through represented physical conditions

CQ2 — Candidate Identification and Comparison	Multiple Assessments, FeatureValue, requirement satisfaction and conflict relations	Identify and compare candidates under the modeled constraints
CQ3 — Alternative Configuration Comparison	ObjectPart, hasPart, isPartOf, interactsWith, part-level FeatureValue	Compare configurations involving different part structures and physical characteristics